

"Single nuclei ATAC sequencing of postmortem cingulate cortex, midbrain and motor cortex of healthy donors and Parkinson's disease patients – Scale-ATAC + 10x Genomics."

Dataset Description:

This dataset consists of raw sequencing ATAC-seq data (Scale-ATAC pre-indexing followed by 10x Genomics snATAC v2). The data is part of an overall set of samples derived from postmortem midbrain (n=140), cingulate cortex (n=190) and motor cortex (n=4) of healthy donors (n=114), patients with Parkinson's disease (n=75) or patients with other neurological disorder (n=1). The protocol followed to isolate nuclei from postmortem brain samples and to prepare sequencing libraries can be found here: <https://dx.doi.org/10.17504/protocols.io.14egnr7ql5d/v1>, <https://dx.doi.org/10.17504/protocols.io.4r3l2z24ql1y/v1>, https://cdn.10xgenomics.com/image/upload/v1666737555/support-documents/CG000496_Chromium_NextGEM_SingleCell_ATAC_ReagentKits_v2_UserGuide_RevB.pdf. To increase throughput and to decrease batch effects, several donors have been pooled together into a single sequencing library. To computationally demultiplex the nuclei to their corresponding donors, cell-snp-lite (version commit: aad18644adcde853c313362a856a24245c9b91f7) followed by vireo (<https://github.com/single-cell-genetics/vireo/pull/108>) has been used. The population VCF with the donor genotypes derived from whole genome sequencing data has been used to assign nuclei back to their donors.

Authors:

- Paníková, Alexandra; [ORCID:0000-0002-0693-132X](https://orcid.org/0000-0002-0693-132X); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium, & Laboratory of Integrative Cancer Genomics, VIB-KU Leuven Center for Cancer Biology, Leuven, Belgium, & Department of Oncology, KU Leuven, Leuven, Belgium
- Theunis, Koen; [ORCID:0000-0002-0699-6676](https://orcid.org/0000-0002-0699-6676); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium
- Hulselmans, Gert; [ORCID:0000-0003-2205-1899](https://orcid.org/0000-0003-2205-1899); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium
- Sigalova, Olga; [ORCID:0000-0001-8598-1079](https://orcid.org/0000-0001-8598-1079); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium
- De Man, Julie; [ORCID:0009-0003-7208-8961](https://orcid.org/0009-0003-7208-8961); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium
- Voet, Thierry; [ORCID:0000-0003-1204-9963](https://orcid.org/0000-0003-1204-9963); Department of Human Genetics, KU Leuven, Leuven, Belgium, & KU Leuven Institute for Single Cell Omics (LISCO), University of Leuven, KU Leuven, Leuven, Belgium
- Aerts, Stein; [ORCID:0000-0002-8006-0315](https://orcid.org/0000-0002-8006-0315); Laboratory of Computational Biology, VIB Center for AI & Computational Biology (VIB.AI), Leuven, Belgium, & VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium, & Department of Human Genetics, KU Leuven, Leuven, Belgium

ASAP Team: Voet

Dataset Name: voet-pmdbs-sn-atacseq-scalebio-10x, v1.0

Principal Investigator: Stein Aerts stein.aerts@kuleuven.be

Dataset Submitter: Gert Hulselmans gert.hulselmans@kuleuven.be

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ASAP Project: Understanding inherited and acquired genetic variation in Parkinson's disease through single-cell multi-omics analyses: a unique data resource

Project Description: The functional roles of inherited and acquired genetic variation in the pathogenesis of Parkinson's disease (PD) remain largely unknown. Here, we will first study how germline genetic variants at PD-risk loci, which were identified by genome-wide association study (GWAS), perturb the expression of genes in specific cell (sub)populations of the brain and gut. To this aim, we will apply single-cell gene-expression and open-chromatin quantitative trait locus (QTL) analyses, enabling identification of PD-relevant genes and cell (sub)types. We will deliver the mechanisms of PD-candidate gene expression (dys)regulation in the normal condition, with ageing and in PD, as well as a unique single-cell multi-omic resource for the community. Second, we will study the nature and role of somatic mutations in brain and gut cells in PD-etiopathology. Finally, we will characterize biochemical and phenotypic effects of loss- or gain-of-function QTLs and somatic mutations of candidate genes in in vitro and in vivo model systems, including their impact on the neuro-immune axis.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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